
Helicopter Emergency Rescue (HER25)

Executive Summary

AERO60004 - Aeronautics Group Design Project 2025



HALO

Abstract

HEMS aircraft are currently adaptations of existing helicopter models which, whilst adequate, are limited in their ability to serve specific needs of the HEMS mission. The HALO project presents a novel design in the 3500 t class that utilises coaxial rotors for lift, capable of cruising at 140 kts with a range of over 600 km. Powered by twin Honeywell HTS9000 engines, HALO includes advanced aerodynamic features such as active flow control to minimise fuel burn and improves cruise speed. The H tail allows for rear loading and best-in-class turnaround times, whilst the onboard electronics and air conditioning provides for up to 35 kW of auxiliary power for clinical equipment. Advances stability augmentation allows for a smooth ride for clinicians and patients alike, with gust correction and multichannel tracking.

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HER02 Aerodynamics

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Daniel Cooper	Free Wake Vortex Lattice Modelling for Coaxial Rotor Performance Analysis
Zhaotong Jim Liu	Computational Design and Aerodynamic Optimization of a Helicopter Configuration
Wenyue Summer Shao	Coaxial Rotor Sizing and Blade Dynamic Response Analysis
Chiaw Soong Yeo	Optimising Aerodynamic Design of a Coaxial Helicopter Empennage and Fuselage

HER03 Structural Design

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Tristan Chabanel	Landing Gear Design
Mercury Pratt	Design of Helicopter Lifting Surfaces
Wanning Xue	Hub Design and Pitch Control Mechanism

HER04 Systems

Sayed Bin Hannan	Mission Equipment and Sub-systems Design for HEMS Rotorcraft
Sabina Hassan	Propulsion Integration and Performance Prediction for Twin-Engine HEMS Aircraft
Ash Robson	Weight and Balance modelling, System Placement, and Avionics Design of a Coaxial HEMS Rotorcraft
Dylan Tan	Integrated Propulsion and Transmission Design for a Coaxial Helicopter with Active Flow Control

HER05 Flight Mechanics, Performance

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Hanchen Li	Flight Dynamics Modelling, Trim Analysis, Basic Stability Augmentation and Autopilot Design
Haoran Song	Preliminary Performance Estimation

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1 Introduction

HALO is a next-generation coaxial twin-engine helicopter developed for Helicopter Emergency Medical Services (HEMS) operations in both urban and remote environments. The design prioritises compactness, high hover efficiency, and rapid response capability—key attributes required for safe rooftop landings and constrained rescue zones. Developed through a multidisciplinary approach, HALO integrates advanced aerodynamic modelling, structural optimisation, flight control logic, and mission-driven systems engineering.

HALO had a three-tiered variant strategy: Lite, Base, and Plus. Each variant offers a distinct balance of system complexity, performance range, and cost, tailored to different mission profiles and operating regions. For example, the Plus variant integrates advanced systems such as active flow control and extended avionics, while the Lite variant strips back to essential capability for cost-sensitive markets. A combined approach using parametric and regression-based modelling enabled pricing at \$7.4 million (Lite), \$8.18 million (Base), and \$9 million (Plus) - positioning HALO competitively against existing platforms while maintaining high technical ambition.

The following report outlines the integrated engineering solutions behind HER25’s development, from rotor and fuselage design through systems architecture, propulsion, controls, and lifecycle cost modelling. Each section demonstrates how performance, manufacturability, certification feasibility, and operational viability were balanced to deliver a helicopter that is not only aerodynamically refined but also commercially and clinically fit for purpose.

1.1 Flight Envelope and Performance Metrics

The helicopter’s flight envelope was characterised through maximum roll, pitch, and yaw rates based on mission safety margins. Boundary diagrams for CG offset, autorotation, and gust tolerance were developed to ensure operational robustness.

Performance analysis at MTOW and no-payload configurations evaluated both point and mission-level metrics at cruise altitude. Key results are shown in Tables 1 and 2, including range, endurance, turn radius, and climb rates.

Table 1 Optimum Performance

	Ceiling (m)	Hover Ceiling (m)	Rate of Descent (m/s)	Max Forward Speed (m/s)
No payload	12980	8500	5.47	87.3
MTOW	10850	6040	6.79	86.4
	Max Range (km)	Max Endurance (h)	Rate of Climb (m/s)	
No payload	1149	8.3	16.5	
MTOW	1018	6.6	12.0	

Table 2 Cruise Performance

Range (km)	Endurance (h)	Min turn radius-MTOW (m)	Min turn radius-No payload (m)
757	2.9	884	442

2 Rotor Design

A coaxial rotor configuration was selected to optimise footprint, hover efficiency, and urban operability. Trade-off analyses determined a dual three-bladed system with a tip speed of 220 m/s and an aspect ratio of 20. Mass convergence was achieved using NASA’s Aeroflightdynamics Directorate (AFDD) weight model 82. Given the design MTOW of 3500 kg and a rotor diameter of 10.61 m (with 20%

root cutout), rotor twist and taper were optimised using Blade Element Momentum Theory (BEMT), accounting for coaxial interference and wake interactions.

Early-stage development used Blade Element Momentum Theory (BEMT) for rapid rotor performance evaluation, but limitations in modelling inter-rotor interference prompted the transition to high-fidelity methods. The use of VSPAERO, a free wake vortex lattice solver, enabled accurate prediction of complex wake interactions and guided critical refinements to thrust balance and torque trimming between the upper and lower rotors. Complementary computational fluid dynamics (CFD) work also informed fuselage shaping, blade airfoil selection, and drag mitigation strategies—leading to a 43% reduction in total drag and improved aerodynamic stability.

2.1 Rotor Geometry

Aerofoil selection prioritised high lift, low drag, compressibility tolerance, and low pitching moment. Using XFOIL and CFD, transition regions were refined to ensure smooth lift distribution and wake stability. Dynamic parameters, including flapping frequency, coning angle, and cyclic flapping coefficients, were computed for hover and forward flight, confirming balanced lift and controlled blade deflection. Static stall analysis validated retreating blade stall margins at high advance ratios.

The upper rotor used a higher negative twist ($\theta_{twist} = -11^\circ$), while the lower rotor used a less aggressive twist ($\theta_{twist} = -6^\circ$), achieving balanced performance between hover (figure of merit $FM = 0.472$) and cruise ($L/D = 6.29$). The thrust ratio $C_T^u/C_T^l = 1.24$ provided torque balance and mitigated aerodynamic asymmetry due to wake interaction. Taper was introduced on the upper rotor (taper ratio 1.6 starting at $r = 0.5$) to align blade loading coefficients (C_T/σ_e) and increase stall margins by reducing tip loading.

2.2 High-Fidelity Modelling and CFD Refinement

VSPAERO, a high-fidelity free-wake vortex lattice solver, was used to validate coaxial rotor performance. After verifying against an in-house single-rotor solver, mesh and wake convergence studies confirmed reliability. BEMT accurately predicted upper rotor thrust but over-predicted lower rotor thrust and rotor torques due to wake interference, indicating limitations in torque trim estimation and necessitating higher-fidelity tools for final optimisation.

Fuselage aerodynamics were refined through iterative CFD simulations assessing rear ramp extensions, tail configuration, and local shaping. These changes progressively delayed flow separation and reduced pressure drag. The final design achieved a 43% drag reduction, down to 1999 N, and a lift reduction from 3915 N to 1234 N due to reduced frontal area and smoother wake alignment.

2.3 Rotor Structure

Rotor blades were constructed using high-strength carbon fibres within an epoxy matrix, providing high fatigue life and exceptional specific strength under cyclic loads. Analytical calculations of bending moments and shear forces at the root guided skin and spar thickness sizing. The upper blade required a 2 mm skin and 4 mm spar (12.6 kg), while the lower blade needed 4 mm skin and 6 mm spar (20.6 kg). Final designs were validated via finite element simulations and fatigue analysis.

Lead-lag analysis indicated resonance risk due to the regressing lag frequency aligning with the fuselage's natural frequency. A mechanical lead-lag damper was implemented to mitigate instability.

The rotor hub was comprised of a central hub structure, a 17-4PH stainless steel torque tube (with a 4 mm wall thickness) for torque transmission, and a lexbeam made of epoxy/high-strength carbon fibre UD layup, enabling $\pm 20^\circ$ pitch angles,

Two flexbeam designs, rectangular and cruciform, were evaluated via FEA. The cruciform design reduced torsional stiffness by 45% while maintaining safety factors above 1.5. Modal analysis confirmed natural frequencies (4.7 Hz and 11.3 Hz) were well-separated from rotor excitation frequencies. A coaxial

pitch control system using a servo-actuated lower swashplate enabled precise synchronised control of both rotors.

3 Fuselage Design

A smoothly tapered rear fuselage houses a downward-opening ramp door, enhancing accessibility and reducing wake size. Landing skid cross tubes were partially embedded into the fuselage with fairings to streamline flow. A hub cap was added to reduce interference drag at the rotor hub.

Engine inlets and outlets were aligned with the turboshaft engine for optimal mass flow. An active flow control system was integrated, redirecting exhaust through rear fuselage nozzles angled backward at velocities three times the cruise speed. Activated only during cruise, this system reduced drag by an additional 10%.

3.1 Material Selection

Structural members (frames and stringers) were made from Aluminium 7075-T6, and the skin used a quasi-isotropic layup of HS carbon fibre/epoxy prepreg. This combination balances lightweight construction with material cost. Analytical sizing of members and skins was validated through finite element simulations under hover conditions. Tail booms were sized analytically using cantilever beam theory to resist self-weight and tailplane loads. The full fuselage structure: including cabin, cockpit, floor, skin, and tail booms—was estimated at 355 kg.

3.2 Empennage Design

To reduce download and overall drag, a twin-boom-supported H-tail was adopted. All-moving vertical and horizontal tailplanes pivot about a circular spar placed at 30% chord of the NACA 0015 aerofoil.

Tail structures mirrored the rotor blade material and internal layout but required thinner dimensions due to lower aerodynamic loads. Vertical and horizontal tails used 1 mm skin thicknesses, with spar thicknesses of 2 mm and 1.5 mm, respectively. Masses were 6 kg (vertical) and 16.6 kg (horizontal), totalling 28.6 kg.

3.3 Landing Gear Design

Skid-type landing gear was selected for terrain versatility, low weight, and simplicity. Constructed from Al 7068-T6511 (yield stress: 702 MPa), the gear was sized using beam theory and refined via FEA to meet CS-29 hard landing requirements. Final dimensions and curvature are shown in Table 3. Total landing gear mass is 78 kg. Modal analysis showed the first three vibrational modes at 13.6 Hz, 20.4 Hz, and 39.0 Hz.

Table 3 Final Landing Gear Dimensions

Component	Straight Section Length (m)	Diameter (mm)	Thickness (mm)	Height (m)	Spacing (m)	Curvature, κ (1/m)
Skid Tube	3.7	80 (H), 120 (V)	8	—	2.0	—
Cross Tube	0.9	80	12	0.3	2.6	2.6

4 Propulsion & Systems

4.1 Powertrain and Transmission

Following a comparative evaluation of piston, electric, hybrid, and turboshaft engines, the Honeywell HTS900 turboshaft was selected for HALO. It provides 667 kW of power per engine with a specific fuel consumption of 319.45 g/kWh. The engine features a 0.034 m² inlet area, 0.0349 m² exhaust area, 2.6 kg/s inlet mass flow, and 0.05919 kg/s fuel flow rate. The never-exceed speed of 80.1 m/s maintains a 10% margin over design cruise speed. Twin-engine integration includes inlet filtration to protect against environmental contaminants and provides bleed air for secondary systems.

Power is transmitted via a compact four-stage gearbox using spiral bevel and planetary gears. This configuration yields a 15.8:1 reduction ratio with 92.2% estimated efficiency. Shaft lengths were minimised to reduce mass and volume, with torque-based sizing ensuring reliability. Final transmission weight: 188.8 kg.

4.2 Fuel, Electrical and Thermal Systems

Crash-resistant CFRP fuel tanks are bottom-mounted and split between forward and aft cells. Dual electric boost pumps per tank, check-valved pickups, and a cross-feed valve provide reliable delivery. FADEC-controlled transfer and multi-stage filtration ensure balance and operational integrity. Pressure, flow, and fuel level sensors are connected to cockpit alerts and shutoff valves for rapid isolation in fault conditions.

Two 28 V DC Skurka generators (12 kW each) supply electrical power via a common main bus with fault isolation and load shedding. The total electrical load under maximum demand is 15.75 kW with a 10% margin. In OEI scenarios, one generator maintains 9.94 kW of essential loads. Dual True Blue lithium-ion batteries provide 30 minutes of backup power (4.44 kW minimum) to support safe landing in full electrical emergency.

Battery capacity totals 2.64 kWh, with passive/active thermal management. Power electronics and all medical systems and inverters are modularly rack-mounted for space efficiency and cooling.

4.3 De-Icing, Drag Reduction and Weight Modelling

HALO implements a three-tier de-icing strategy tailored to each icing hazard. The engine inlets are equipped with electrothermal strips drawing 2.85 kW to prevent ice ingestion and flameouts. The windscreen is protected by a bleed-air system that maintains clear visibility without optical distortion. The rotor blades and tail surfaces are fitted with three 40 kHz piezoelectric actuators per blade, located at A_0 Lamb-wave quarter-wavelength antinodes. These actuators generate shear forces sufficient to fracture accreted ice at an energy density of 890 W/m². This ultrasonic de-icing system requires only 2.52 kW across both rotors, achieving an 82% power saving over conventional electrothermal systems, with the total de-icing load limited to 4.957 kW.

The Active Flow Control System (AFCS) adds 26.4 kg and is non-critical, meaning its failure does not compromise operational safety, making it a low-risk, high-reward enhancement.

A lumped mass model was used to define the centre of gravity across all loading configurations. The final layout results in an empty weight of 1747.4 kg, a mission-ready weight of 3333.4 kg, and an MTOW of 3500 kg. System placement was automatically optimised during design using a custom algorithm to achieve the target centre of gravity and maintain balance across flight conditions.

5 Avionics and Control

5.1 Avionics, Controls and Safety

HALO uses a glass cockpit with primary, secondary, navigation, and standby displays, integrated with HTAWS and NVIS systems. A FLIR pod aids night navigation and landing site selection. Avionics

follow the ARINC 429 standard, allowing sensor modularity for client-specific upgrades.

Flight controls are hydraulically boosted via dual redundant loops, enabling both digital control and mechanical override in failure modes (see Section 5.2).

Safety systems include triplex flight data computers, HTAWS, WSPS, and full FADEC with digital twins for predictive maintenance. Redundancy covers all critical subsystems. Future upgrades include a thermoelectric generator (TEG) in the exhaust stream to power essential systems during complete generator failure.

5.2 Control Architecture and Stability Analysis

HALO employs cyclic, collective, and differential rotor control strategies, tailored to maintain coaxial symmetry. Manual override and stability augmentation are included for medical/emergency operations.

The system operates across five control modes: Attitude Command Attitude Hold (ACAH), Rate Command Direction Hold (RCDH), Altitude Hold (AH), Translational Rate Command (TRC), and Rate Damping (RD), the latter serving as the inner loop to suppress high-frequency oscillations. A triple-cascade architecture with PID baseline control was implemented, and an H_∞ controller was trialled to improve robustness, though it could not satisfy robustness and performance simultaneously.

A linearised 6-DOF model supported hover and trim stability analysis. Root locus and Nyquist techniques verified dynamic stability. Pilot input mappings were validated through MATLAB simulations to ensure actuator fidelity and response time alignment.